

Order Araneae

Spiders

Spiders can be distinguished from other arachnids by their distinctive appearance and by their ability to produce silk. A typical spider has 8 eyes, and a body divided into 2 parts—the cephalothorax and the abdomen. The mouth is flanked by a pair of venom-injecting fangs, and a pair of leglike pedipalps. These have a sensory function and, in males, are also used to transfer sperm. All spiders are predatory, injecting venom into their prey. Some use silk to catch prey, but spiders also use silk to protect their eggs, lower themselves through the air, and even to sail on the breeze. There are over 42,000 species, in about 110 families, and they live in every terrestrial habitat, from tropical rain forests to cellars and caves.

Family Araneidae

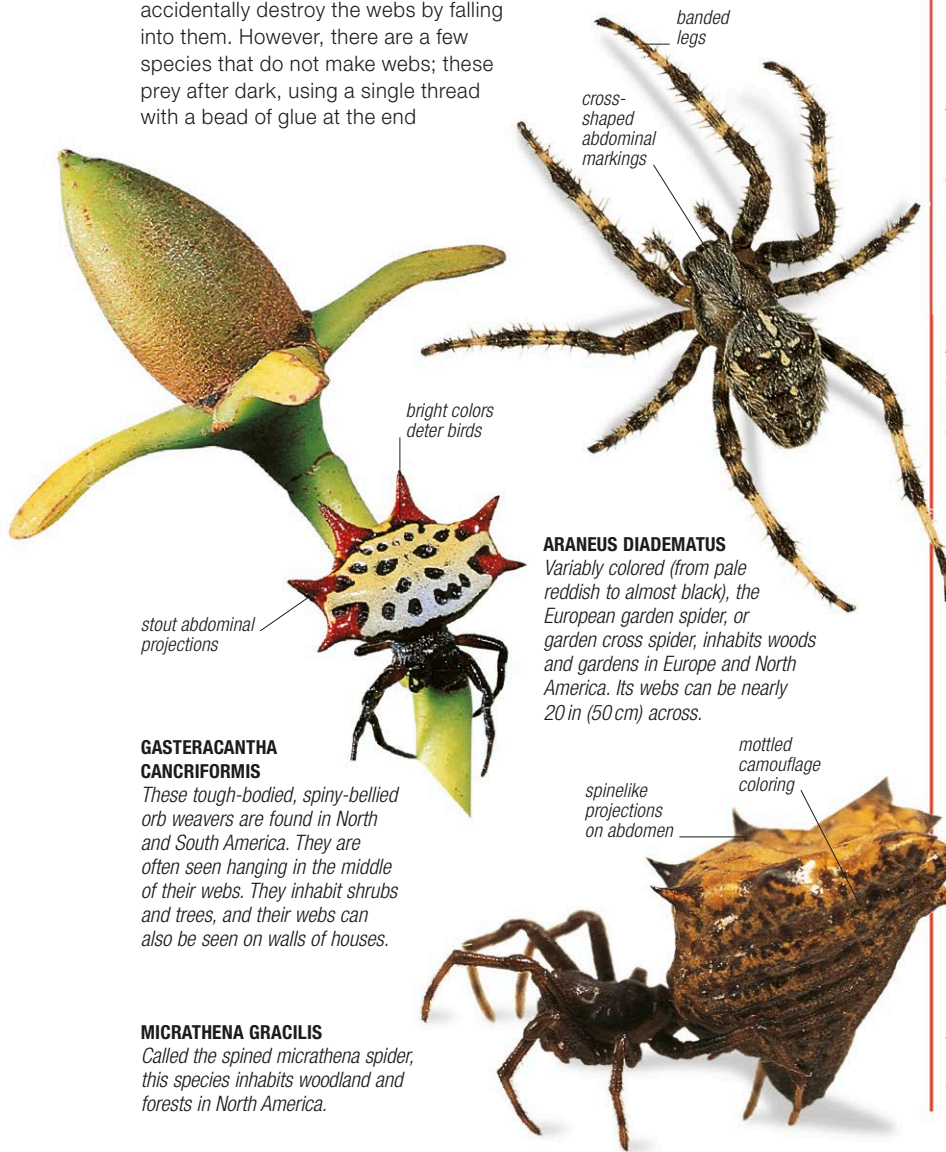
Orb web spiders

Occurrence c.3,100 spp. worldwide; in grassland, meadows, woodland, forests, gardens

Habitat Terrestrial

As the name implies, many of these spiders weave circular webs made up of a central hub with radiating lines and spirals of sticky and nonsticky silk. They typically catch prey in the web, wrap it in silk, cut it out, and take it to a retreat to eat it. Some species spin opaque bands of matted silk in their webs so that birds can see them and do not accidentally destroy the webs by falling into them. However, there are a few species that do not make webs; these prey after dark, using a single thread with a bead of glue at the end

to snare moths. The strength of the webs can be such that in Papua New Guinea, the huge webs of the *Nephila* species are used as fishing nets. Orb web spiders often have very large, brightly colored, and patterned abdomens; in tropical species, the abdomen may have unusual, angular shapes. The legs have 3 claws that can be very spiny. Of the 8 eyes, the 4 in the middle often form a square. Orb web spiders are 1/16–1 3/4 in (0.2–4.6 cm) long. Males approach females on their webs and pluck threads to attract them. Females lay eggs in silk cocoons that can be camouflaged in webs and litter, or stuck to vegetation or bark.



GASTERACANTHA CANCRIFORMIS

These tough-bodied, spiny-bellied orb weavers are found in North and South America. They are often seen hanging in the middle of their webs. They inhabit shrubs and trees, and their webs can also be seen on walls of houses.

MICRATHENA GRACILIS

Called the spined micrathena spider, this species inhabits woodland and forests in North America.

ARANEUS DIADEMATUS

Variably colored (from pale reddish to almost black), the European garden spider, or garden cross spider, inhabits woods and gardens in Europe and North America. Its webs can be nearly 20 in (50 cm) across.

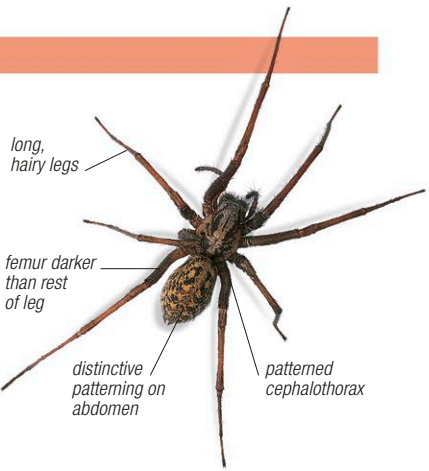
Family Agelenidae

Funnel weavers

Occurrence 1,150 spp. worldwide; in meadows, gardens, bushes, houses, among stones, on walls



Funnel weavers are often long-legged, and have minute, feathery hairs on the body, 8 eyes, and a slender abdomen with dark bars, chevrons, or spots. The body is 1/4–3/4 in (0.6–2 cm) long. The funnel weaver's web takes the form of a flat, tangled silk sheet with a tunnel on one side. Females may feed their offspring by regurgitation and, in some species, the spiderlings may eat the mother when she dies.



TEGENARIA DUELICA

This giant house spider can have a legspan of up to 3 1/4 in (8 cm), and can run very fast. Members of the genus Tegenaria are called house spiders. The large spiders that fall into domestic baths usually belong to this genus.

Family Cybaeidae

Cybaeid spiders

Occurrence 180 species in North America, South America, and Eurasia



Over 100 species of *Cybaeus* make webs in stone crevices or under logs, but this family's best known member is the water spider (*Argyroneta*): the only spider that lives under water. It ferries air, trapped by its dense abdominal hair, to a silken "diving bell" anchored to pond weed. The bell is used for feeding, courtship, hibernation, and stowing eggs.



ARGYRONETA AQUATICA

This spider lives permanently in still or slow-moving water, and the air trapped by the body hairs around the abdomen gives it a silvery appearance.

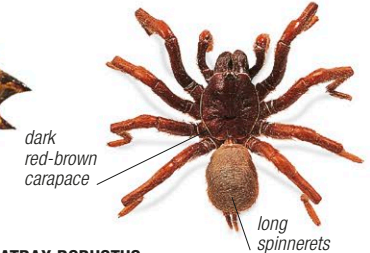
Family Hexathelidae

Funnel-web spiders

Occurrence 180 spp. in tropical regions of the Americas, Africa, Asia, and Australia; in trees, on ground



Funnel-web spiders belong to a group known as mygalomorphs, which also includes tarantulas (see p.593). Unlike other spiders, mygalomorphs are relatively primitive, with fangs that strike downward, instead of closing sideways like a pair of pincers. Funnel-webs are 3/8–1 1/4 in (1–3 cm) long and, usually, dark brown. Most make a funnel-shaped retreat that leads into crevices in tree stumps or rocks. Some are highly venomous.



ATRAX ROBUSTUS

The Sydney funnel-web, a native of Southeast Australia, is an aggressive spider that has a painful, venomous bite that may prove fatal.

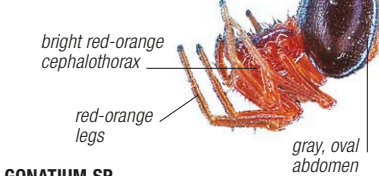
Family Linyphiidae

Dwarf spiders

Occurrence 4,400 spp. mostly in temperate regions of the north, some in Arctic; on stones, among vegetation



Also known as money spiders, these tiny animals are rarely more than 3/8 in (1 cm) long. They have large chelicerae with sharp teeth. The legs have strong bristles and the carapace of males may have extensions, which sometimes carry the eyes. Dwarf spiders live in leaf litter or make characteristic, horizontal, nonsticky, sheetlike webs among vegetation. Egg sacs are attached to stones, plants, and other surfaces. These spiders travel over long distances by "ballooning" on long, silk threads.



GONATIUM SP.

These small spiders are widespread in the Northern Hemisphere, inhabiting scrubland and grassland. They prefer low, shaded bushes.